Sec 1

Directions for question 1 to 4: Answer the questions on the basis of information given below.

Six persons namely Andy, Bharat, Chahel, Daksh, Ekta and Faiz played a game in which they had to guess the prices of five different objects A, B, C, D and E. Later it was found that each of them guessed the price of exactly one object correctly. Also, the price of each item was guessed correctly by at least one person. For each object one guess was half the original price and one guess was double the original price. The table given below shows the prices (in Rs.) guessed by these six persons for each object:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14	56	28	20	7	22
B	9	13	36	52	26	18
C	12	24	6	3	48	10
D	36	16	72	18	9	36
E	5	30	15	60	120	30

Q.1 [11831809]

What is the price (in Rs.) of the object guessed correctly by Ekta?

Solution:

Correct Answer : 26

Answer key/Solution

As each person guessed correctly the price of exactly one object, it can happen only when two persons guess the price of one object correctly. From the table we can see that two objects for which the prices was correctly guessed by two persons can either be D or E i.e., when D's price is Rs. 36 or E's price is Rs. 30.

For each object one guess was half the original price and one guess was double the original price, so the possible prices (in Rs.) can be:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14		28			
B					26	
C	12	24	6			
D	36			18		36
E		30		60		30

Ekta guessed the price of object B correctly i.e., Rs. 26.

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Directions for question 1 to 4: Answer the questions on the basis of information given below.

Six persons namely Andy, Bharat, Chahel, Daksh, Ekta and Faiz played a game in which they had to guess the prices of five different objects A, B, C, D and E. Later it was found that each of them guessed the price of exactly one object correctly. Also, the price of each item was guessed correctly by at least one person. For each object one guess was half the original price and one guess was double the original price. The table given below shows the prices (in Rs.) guessed by these six persons for each object:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14	56	28	20	7	22
B	9	13	36	52	26	18
C	12	24	6	3	48	10
D	36	16	72	18	9	36
E	5	30	15	60	120	30

Q.2 [11831809]

If the price of object D was not Rs. 18, then what was the price (in Rs.) of the object C?

Solution:

Correct Answer : 24

Answer key/Solution

As each person guessed correctly the price of exactly one object, it can happen only when two persons guess the price of one object correctly. From the table we can see that two objects for which the prices was correctly guessed by two persons can either be D or E i.e., when D's price is Rs. 36 or E's price is Rs. 30.
For each object one guess was half the original price and one guess was double the original price, so the possible prices (in Rs.) can be:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14		28			
B					26	
C	12	24	6			
D	36			18		36
E		30		60		30

If price of object D is not Rs. 18, then it has to be Rs. 36. So Andy and Faiz guessed the price of object D correctly. So price of A cannot be Rs. 14 as each person guessed correctly the price of exactly one object.

So price of object A is Rs. 28, guessed by Chahel.
Now, price of object C cannot be Rs. 6 or Rs.12, so price of object C is Rs. 24.

Bookmark FeedBack

Directions for question 1 to 4: Answer the questions on the basis of information given below.

Six persons namely Andy, Bharat, Chahel, Daksh, Ekta and Faiz played a game in which they had to guess the prices of five different objects A, B, C, D and E. Later it was found that each of them guessed the price of exactly one object correctly. Also, the price of each item was guessed correctly by at least one person. For each object one guess was half the original price and one guess was double the original price. The table given below shows the prices (in Rs.) guessed by these six persons for each object:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14	56	28	20	7	22
B	9	13	36	52	26	18
C	12	24	6	3	48	10
D	36	16	72	18	9	36
E	5	30	15	60	120	30

Q.3 [11831809]

What could be the minimum possible total price (in Rs.) of all the objects taken together?

1 ◯ 86

2 ◯ 100

3 ◯ 108

4 ◯ 94

Solution:
Correct Answer : 4

Answer key/Solution

As each person guessed correctly the price of exactly one object, it can happen only when two persons guess the price of one object correctly. From the table we can see that two objects for which the prices was correctly guessed by two persons can either be D or E i.e., when D's price is Rs. 36 or E's price is Rs. 30.
For each object one guess was half the original price and one guess was double the original price, so the possible prices (in Rs.) can be:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14		28			
B					26	
C	12	24	6			
D	36			18		36
E		30		60		30

Minimum price of all the objects taken together can be Rs. 94, when
price of A = Rs. 14
price of B = Rs. 26
price of C = Rs. 6
price of D = Rs. 18
price of E = Rs. 30
Hence, option (4) is the correct answer.

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Directions for question 1 to 4: Answer the questions on the basis of information given below.

Six persons namely Andy, Bharat, Chahel, Daksh, Ekta and Faiz played a game in which they had to guess the prices of five different objects A, B, C, D and E. Later it was found that each of them guessed the price of exactly one object correctly. Also, the price of each item was guessed correctly by at least one person. For each object one guess was half the original price and one guess was double the original price. The table given below shows the prices (in Rs.) guessed by these six persons for each object:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14	56	28	20	7	22
B	9	13	36	52	26	18
C	12	24	6	3	48	10
D	36	16	72	18	9	36
E	5	30	15	60	120	30

Q.4 [11831809]
If a, b, c, d and e represent the prices (in Rs.) of objects A, B, C, D and E respectively, then which of the following could be true?

1 ◯ a + d + e = 110

2 ◯ b + c + d = 74

3 ◯ a + b + c = 60

4 ◯ a + b + d = 90

Solution:
Correct Answer : 4

Answer key/Solution

As each person guessed correctly the price of exactly one object, it can happen only when two persons guess the price of one object correctly. From the table we can see that two objects for which the prices was correctly guessed by two persons can either be D or E i.e., when D's price is Rs. 36 or E's price is Rs. 30.
For each object one guess was half the original price and one guess was double the original price, so the possible prices (in Rs.) can be:

	Andy	Bharat	Chahel	Daksh	Ekta	Faiz
A	14		28			
B					26	
C	12	24	6			
D	36			18		36
E		30		60		30

Option (1) is possible when a = 14, d = 36 and e = 60; but d = 36 and a = 14 is simultaneously not possible.
Option (2) is possible when b = 26, c = 12 and d = 36; but c = 12 and d = 36 is simultaneously not possible.
Option (3) is possible when a = 28, b = 26 and c = 6; but a = 28 and c = 6 is simultaneously not possible.
Option (4) is possible when a = 28, b = 26 and d = 36.
Hence, option (4) is the correct answer.

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Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

An engineering college receives scholarship funds every year. Then college conducts a test and interview to shortlist certain candidates who receives scholarship as per their performance. The candidate with the highest cumulative score receives the highest scholarship, the candidate with second highest scores receives second highest scholarship, the candidate with third highest scores receives third highest scholarship and so on. There were no ties in the scores of the candidates.
All the shortlisted candidates pay a registration amount of Rs. 20,000 after appearing for the scholarship test. Exactly top 5 candidates receive scholarships such that if a candidate fail to secure more than 15% of the scholarship funds, they lose their registration amount.
The table given below provides some incomplete information about scholarship funds allocated to the shortlisted candidates in four consecutive years 2018 to 2021.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)				1,50,000
Scholarship to candidate with 4th highest score (in Rs.)				10%
Scholarship to candidate with 5th highest score (in Rs.)				

The following additional facts are known:
(i) The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.
(ii) None of the candidates shortlisted in 2020 lost their joining amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000.
(iii) The candidate with second highest score in 2021 received 10% of total scholarship fund more than that of candidate with 3rd highest score, and candidate with highest score received 20% of total scholarship more than that of candidate with 2nd highest score.
(iv) All the candidates receive scholarships in the integral multiples of 10.

Q.5 [11831809]
What is the percentage of scholarship funds allocated to candidates who lost their joining amounts in 2018?

1 ◯ 15%

2 ◯ 10%

3 ◯ 33.33%

4 ◯ 20%

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
From condition (i), The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000			1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	a			10%
Scholarship to candidate with 5th highest score (in Rs.)	0.25a			

Step 2:
So, $1.25a = 100000 - 900000 = 100000 \Rightarrow a$ = Rs. 80,000.
From condition (ii), None of the candidates shortlisted in 2020 lost their registration amount, which implies all the candidates must have received the scholarship amount of more than 15% of the total amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000 which is only possible in the below given scenario.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	

Step 3:
From condition (iii), we get the final table.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	25,000

Two candidates in 2018 received less than 15% of scholarship funds.
Hence, their total amount = Rs. 1,00,000 which is equivalent to 10% of the total scholarship funds.

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Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

An engineering college receives scholarship funds every year. Then college conducts a test and interview to shortlist certain candidates who receives scholarship as per their performance. The candidate with the highest cumulative score receives the highest scholarship, the candidate with second highest scores receives second highest scholarship, the candidate with third highest scores receives third highest scholarship and so on. There were no ties in the scores of the candidates.
All the shortlisted candidates pay a registration amount of Rs. 20,000 after appearing for the scholarship test. Exactly top 5 candidates receive scholarships such that if a candidate fail to secure more than 15% of the scholarship funds, they lose their registration amount.
The table given below provides some incomplete information about scholarship funds allocated to the shortlisted candidates in four consecutive years 2018 to 2021.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)				1,50,000
Scholarship to candidate with 4th highest score (in Rs.)				10%
Scholarship to candidate with 5th highest score (in Rs.)				

The following additional facts are known:
(i) The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.
(ii) None of the candidates shortlisted in 2020 lost their joining amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000.
(iii) The candidate with second highest score in 2021 received 10% of total scholarship fund more than that of candidate with 3rd highest score, and candidate with highest score received 20% of total scholarship more than that of candidate with 2nd highest score.
(iv) All the candidates receive scholarships in the integral multiples of 10.

Q.6 [11831809]
In 2019, all five candidates received scholarship funds which formed Arithmetic Progression, then how many candidates received their registration amount back?

1 ○ 0

2 ◯ 1

3 ◯ 2

4 ◯ 3

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
From condition (i), The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000			1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	a			10%
Scholarship to candidate with 5th highest score (in Rs.)	0.25a			

Step 2:
So, $1.25a = 100000 - 900000 = 100000 \Rightarrow a =$ Rs. 80,000.
From condition (ii), None of the candidates shortlisted in 2020 lost their registration amount, which implies all the candidates must have received the scholarship amount of more than 15% of the total amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000 which is only possible in the below given scenario.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	

Step 3:
From condition (iii), we get the final table.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	25,000

In 2019, if all five candidates received scholarship funds which formed Arithmetic Progression, then the only possibility is:

	College Scholarships 2018-2021			
	2018	2019	2020	2021

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No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000	5,00,000	2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,005	3,00,000	1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000	2,00,000	1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000	1,00,000	1,20,010	25,000

Since 15% of Rs. 15,00,000 is Rs. 2,25,000, therefore, three candidates must have received their registration amount back.

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

An engineering college receives scholarship funds every year. Then college conducts a test and interview to shortlist certain candidates who receives scholarship as per their performance. The candidate with the highest cumulative score receives the highest scholarship, the candidate with second highest scores receives second highest scholarship, the candidate with third highest scores receives third highest scholarship and so on. There were no ties in the scores of the candidates.

All the shortlisted candidates pay a registration amount of Rs. 20,000 after appearing for the scholarship test. Exactly top 5 candidates receive scholarships such that if a candidate fail to secure more than 15% of the scholarship funds, they lose their registration amount.

The table given below provides some incomplete information about scholarship funds allocated to the shortlisted candidates in four consecutive years 2018 to 2021.

	College Scholarships 2018-2021			
	2018	**2019**	**2020**	**2021**
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)				1,50,000
Scholarship to candidate with 4th highest score (in Rs.)				10%
Scholarship to candidate with 5th highest score (in Rs.)				

The following additional facts are known:

(i) The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

(ii) None of the candidates shortlisted in 2020 lost their joining amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000.

(iii) The candidate with second highest score in 2021 received 10% of total scholarship fund more than that of candidate with 3rd highest score, and candidate with highest score received 20% of total scholarship more than that of candidate with 2nd highest score.

(iv) All the candidates receive scholarships in the integral multiples of 10.

Q.7 [11831809]

What BEST can be concluded about the scholarship amount (in Rs.) received by the candidate with highest score in year 2020?

1 ◯ between 1,50,000 and 1,80,050

2 ◯ 2,00,010

3 ◯ 1,80,020

4 ◯ less than 1,95,000

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
From condition (i), The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000			1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	a			10%
Scholarship to candidate with 5th highest score (in Rs.)	0.25a			

Step 2:
So, $1.25a = 100000 - 900000 = 100000 \Rightarrow a$ = Rs. 80,000.
From condition (ii), None of the candidates shortlisted in 2020 lost their registration amount, which implies all the candidates must have received the scholarship amount of more than 15% of the total amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000 which is only possible in the below given scenario.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	

Step 3:
From condition (iii), we get the final table.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	25,000

The scholarship amount received by the candidate with highest score in year 2020 is Rs. 2,00,010.

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Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

An engineering college receives scholarship funds every year. Then college conducts a test and interview to shortlist certain candidates who receives scholarship as per their performance. The candidate with the highest cumulative score receives the highest scholarship, the candidate with second highest scores receives second highest scholarship, the candidate with third highest scores receives third highest scholarship and so on. There were no ties in the scores of the candidates.
All the shortlisted candidates pay a registration amount of Rs. 20,000 after appearing for the scholarship test. Exactly top 5 candidates receive scholarships such that if a candidate fail to secure more than 15% of the scholarship funds, they lose their registration amount.
The table given below provides some incomplete information about scholarship funds allocated to the shortlisted candidates in four consecutive years 2018 to 2021.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)				1,50,000
Scholarship to candidate with 4th highest score (in Rs.)				10%
Scholarship to candidate with 5th highest score (in Rs.)				

The following additional facts are known:
(i) The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.
(ii) None of the candidates shortlisted in 2020 lost their joining amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000.
(iii) The candidate with second highest score in 2021 received 10% of total scholarship fund more than that of candidate with 3rd highest score, and candidate with highest score received 20% of total scholarship more than that of candidate with 2nd highest score.
(iv) All the candidates receive scholarships in the integral multiples of 10.

Q.8 [11831809]
What was the scholarship fund (in Rs.) received by the 5th highest scorer in 2021?

Solution:

Correct Answer : 25000

Answer key/Solution

Step 1:

From condition (i), The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000			1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	a			10%
Scholarship to candidate with 5th highest score (in Rs.)	0.25a			

Step 2:

So, $1.25a = 100000 - 900000 = 100000 \Rightarrow a =$ Rs. 80,000.

From condition (ii), None of the candidates shortlisted in 2020 lost their registration amount, which implies all the candidates must have received the scholarship amount of more than 15% of the total amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000 which is only possible in the below given scenario.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	

Step 3:

From condition (iii), we get the final table.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	25,000

The scholarship fund received by the 5th highest scorer in 2021 is Rs. 25,000.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

An engineering college receives scholarship funds every year. Then college conducts a test and interview to shortlist certain candidates who receives scholarship as per their performance. The candidate with the highest cumulative score receives the highest scholarship, the candidate with second highest scores receives second highest scholarship, the candidate with third highest scores receives third highest scholarship and so on. There were no ties in the scores of the candidates.
All the shortlisted candidates pay a registration amount of Rs. 20,000 after appearing for the scholarship test. Exactly top 5 candidates receive scholarships such that if a candidate fail to secure more than 15% of the scholarship funds, they lose their registration amount.
The table given below provides some incomplete information about scholarship funds allocated to the shortlisted candidates in four consecutive years 2018 to 2021.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)				1,50,000
Scholarship to candidate with 4th highest score (in Rs.)				10%
Scholarship to candidate with 5th highest score (in Rs.)				

The following additional facts are known:
(i) The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.
(ii) None of the candidates shortlisted in 2020 lost their joining amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000.
(iii) The candidate with second highest score in 2021 received 10% of total scholarship fund more than that of candidate with 3rd highest score, and candidate with highest score received 20% of total scholarship more than that of candidate with 2nd highest score.
(iv) All the candidates receive scholarships in the integral multiples of 10.

Q.9 [11831809]
In 2019, all five candidates received scholarship funds which formed Arithmetic Progression, then total how much registration amount (in Rs.) remained with the college at the end of the scholarship allocation from 2018 to 2021?

Solution:

Correct Answer : 440000

Answer key/Solution

Step 1:

From condition (i), The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000			1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	a			10%
Scholarship to candidate with 5th highest score (in Rs.)	0.25a			

Step 2:

So, $1.25a = 100000 - 900000 = 100000 \Rightarrow a$ = Rs. 80,000.

From condition (ii), None of the candidates shortlisted in 2020 lost their registration amount, which implies all the candidates must have received the scholarship amount of more than 15% of the total amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000 which is only possible in the below given scenario.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	

Step 3:

From condition (iii), we get the final table.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	25,000

Total candidates every year who either registered and didn't receive any scholarship and who received less

than 15% scholarship = 9 + 5 + 5 + 3 = 22
Hence, required amount = 22 × 20000 = Rs. 4,40,000.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

An engineering college receives scholarship funds every year. Then college conducts a test and interview to shortlist certain candidates who receives scholarship as per their performance. The candidate with the highest cumulative score receives the highest scholarship, the candidate with second highest scores receives second highest scholarship, the candidate with third highest scores receives third highest scholarship and so on. There were no ties in the scores of the candidates.
All the shortlisted candidates pay a registration amount of Rs. 20,000 after appearing for the scholarship test. Exactly top 5 candidates receive scholarships such that if a candidate fail to secure more than 15% of the scholarship funds, they lose their registration amount.
The table given below provides some incomplete information about scholarship funds allocated to the shortlisted candidates in four consecutive years 2018 to 2021.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)				1,50,000
Scholarship to candidate with 4th highest score (in Rs.)				10%
Scholarship to candidate with 5th highest score (in Rs.)				

The following additional facts are known:
(i) The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.
(ii) None of the candidates shortlisted in 2020 lost their joining amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000.
(iii) The candidate with second highest score in 2021 received 10% of total scholarship fund more than that of candidate with 3rd highest score, and candidate with highest score received 20% of total scholarship more than that of candidate with 2nd highest score.
(iv) All the candidates receive scholarships in the integral multiples of 10.

Q.10 [11831809]
Every year the margin between the scholarship amount received by the candidates who scored highest and 2nd highest respectively varied. Which of the following CANNOT be the list of years, in increasing order of this margin?

1 ◯ 2021, 2019, 2018, 2020

2 ◯ 2021, 2018, 2019, 2020

3 ◯ 2019, 2018, 2021, 2020

4 ◯ 2019, 2021, 2018, 2020

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
From condition (i), The 2nd highest scorer had scholarship of Rs. 1,00,000 more than the 3rd highest scorer in year 2018. 5th highest scorer obtained 25% of what fourth highest scorer obtained in year 2018.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000			
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000		
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000			1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	a			10%
Scholarship to candidate with 5th highest score (in Rs.)	0.25a			

Step 2:
So, $1.25a = 100000 - 900000 = 100000 \Rightarrow a =$ Rs. 80,000.
From condition (ii), None of the candidates shortlisted in 2020 lost their registration amount, which implies all the candidates must have received the scholarship amount of more than 15% of the total amount. The difference in the scholarship received by any pair of candidates in this year was at least Rs. 20,000 which is only possible in the below given scenario.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,050	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	

Step 3:
From condition (iii), we get the final table.

	College Scholarships 2018-2021			
	2018	2019	2020	2021
No. of Candidates shortlisted	12	8	10	6
Total Scholarship funds received (in Rs.)	10,00,000	15,00,000	8,00,000	9,50,000
Scholarship to candidate with highest score (in Rs.)	4,50,000		2,00,010	4,35,000
Scholarship to candidate with 2nd highest score (in Rs.)	2,75,000	4,00,000	1,80,010	2,45,000
Scholarship to candidate with 3rd highest score (in Rs.)	1,75,000		1,60,010	1,50,000
Scholarship to candidate with 4th highest score (in Rs.)	80,000		1,40,010	95,000
Scholarship to candidate with 5th highest score (in Rs.)	20,000		1,20,010	25,000

Clearly, margin between first two scorers in year 2021 is greater than 2020, therefore, option (3) is not possible.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

Eight students - A, B, C, D, E, F, G and H are planning to work on 3 projects - P1, P2, and P3. For these 3 projects they form 3 teams - T1, T2 and T3. Team T1 works on project P1, Team T2 works on project P2 and Team T3 works on project P3. For each project to be successful, the students need to work on certain days, as per the requirement of the projects. The table given below shows information about the different days that the students can work for the projects:

Student	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
A	✓	x	x	x	X	x	x	x	✓
B	X	✓	✓	x	X	✓	x	x	X
C	X	✓	✓	✓	✓	x	✓	x	X
D	X	X	x	✓	✓	x	✓	x	X
E	X	X	✓	X	X	x	✓	x	X
F	✓	✓	x	X	X	x	x	x	✓
G	X	X	✓	X	X	x	x	x	✓
H	X	X	✓	X	X	x	✓	✓	X

Following is the additional information:
(i) The number of students in team T1, T2 and T3 are 5, 5 and 4 respectively. Each of the 8 students is in at least one team and being part of the team, must work for at least one day. No student can work for more than two days for a particular project.
(ii) In all the 3 teams, one student works on Day 1, one student works on Day 2 and one student works on Day 9, such that no two teams have same student who works on Day 9. Also, the student who works on Day 1 for a project does not work on any other day for that particular project.
(iii) F works in team T1 on Day 9 and G is in team T3. C is in two teams. Also, C cannot be in the same team as either H or D.
(iv) If work is done on a project on Day 3, then it will be done on Day 5 also. Similarly, if work is done on a project on Day 8, then it will be done on Day 7 also. If D is in a team, he always works for two days on the project. The same is true for G.
(v) In 2 teams out of 3, work is done on Day 6. Similarly, in 2 teams out of 3, work is done on Day 3 and the same is true for Day 4. While on Day 8, one student of only one team works.
(vi) Project P1 requires the students of Team T1 to work on exactly 7 different days out of the 9 different days listed in the table above. Similarly, project P2 requires students of team T2 to work on exactly 7 different days and project P3 requires students of team T3 to work on exactly 5 different days.

Q.11 [11831809]
E is in how many teams?

1 ◯ 1

2 ◯ 2

3 ◯ 3

4 ◯ Either (1) or (2)

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Team T3:
From conditions (iii) and (iv), G works on Day 9 as well as on Day 3.
From condition (iv), If work is done on a project on Day 3, then it will be work on Day 5 also. It has to be C as D cannot work on Day 5. Since D cannot work two days in this team
From conditions (i) and (vi), In team T3, there are given to be 4 members and they work on 5 different days with G works on two days. For Day 1 and Day 2, there are two choices for each. If F works on Day 1, then B works on Day 2 and if A works on Day 1, then on Day 2 the work can be done by either F or B.
Step 2:
Team T1 and Team T2:
From condition (ii), one student works on Day 1, one student works on Day 2 and one student works on Day 9 in each team.
From condition (v), in each of the two teams, one student must work on Day 6, one student on Day 4; while in one of the two teams, from condition (iv), one student must work on Day 3 and hence also one student must work on Day 5.
Also, from condition (iv), In the other team, one student must work on Day 8 and hence also one student must work on Day 7.
Step 3:
For teams T1 and T2, either there is case I for both or there is case II for both.
Hence, the final table can be shown as:

	Team T1		Team T2		Team T3
	Case I	Case II	Case I	Case II	
Day 1	A	A	F	F	F/A
Day 2	B/F	B/F	B	B	F/B
Day 3	E	x	x	E	G
Day 4	C	D	D	C	X
Day 5	C	x	x	C	C
Day 6	B	B	B	B	X
Day 7	x	D	D	x	X
Day 8	x	H	H	x	X
Day 9	F	F	A	A	G

Either E is in team T1 or team T2, but in any case, he is in one team only.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

Eight students - A, B, C, D, E, F, G and H are planning to work on 3 projects - P1, P2, and P3. For these 3 projects they form 3 teams - T1, T2 and T3. Team T1 works on project P1, Team T2 works on project P2 and Team T3 works on project P3. For each project to be successful, the students need to work on certain days, as per the requirement of the projects. The table given below shows information about the different days that the students can work for the projects:

Student	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
A	✓	x	x	x	X	x	x	x	✓
B	X	✓	✓	x	X	✓	x	x	X
C	X	✓	✓	✓	✓	x	✓	x	X
D	X	X	x	✓	✓	x	✓	x	X
E	X	X	✓	X	X	x	✓	x	X
F	✓	✓	x	X	X	x	x	x	✓
G	X	X	✓	X	X	x	x	x	✓
H	X	X	✓	X	X	x	✓	✓	X

Following is the additional information:
(i) The number of students in team T1, T2 and T3 are 5, 5 and 4 respectively. Each of the 8 students is in at least one team and being part of the team, must work for at least one day. No student can work for more than two days for a particular project.
(ii) In all the 3 teams, one student works on Day 1, one student works on Day 2 and one student works on Day 9, such that no two teams have same student who works on Day 9. Also, the student who works on Day 1 for a project does not work on any other day for that particular project.
(iii) F works in team T1 on Day 9 and G is in team T3. C is in two teams. Also, C cannot be in the same team as either H or D.
(iv) If work is done on a project on Day 3, then it will be done on Day 5 also. Similarly, if work is done on a project on Day 8, then it will be done on Day 7 also. If D is in a team, he always works for two days on the project. The same is true for G.
(v) In 2 teams out of 3, work is done on Day 6. Similarly, in 2 teams out of 3, work is done on Day 3 and the same is true for Day 4. While on Day 8, one student of only one team works.
(vi) Project P1 requires the students of Team T1 to work on exactly 7 different days out of the 9 different days listed in the table above. Similarly, project P2 requires students of team T2 to work on exactly 7 different days and project P3 requires students of team T3 to work on exactly 5 different days.

Q.12 [11831809]
Among the following students who could have worked on Day 3 in team T1?

1 ○ B

2 ○ C

3 ○ E

4 ◯ H

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Team T3:
From conditions (iii) and (iv), G works on Day 9 as well as on Day 3.
From condition (iv), If work is done on a project on Day 3, then it will be work on Day 5 also. It has to be C as D cannot work on Day 5. Since D cannot work two days in this team
From conditions (i) and (vi), In team T3, there are given to be 4 members and they work on 5 different days with G works on two days. For Day 1 and Day 2, there are two choices for each. If F works on Day 1, then B works on Day 2 and if A works on Day 1, then on Day 2 the work can be done by either F or B.
Step 2:
Team T1 and Team T2:
From condition (ii), one student works on Day 1, one student works on Day 2 and one student works on Day 9 in each team.
From condition (v), in each of the two teams, one student must work on Day 6, one student on Day 4; while in one of the two teams, from condition (iv), one student must work on Day 3 and hence also one student must work on Day 5.
Also, from condition (iv), In the other team, one student must work on Day 8 and hence also one student must work on Day 7.
Step 3:
For teams T1 and T2, either there is case I for both or there is case II for both.
Hence, the final table can be shown as:

	Team T1		Team T2		Team T3
	Case I	Case II	Case I	Case II	
Day 1	A	A	F	F	F/A
Day 2	B/F	B/F	B	B	F/B
Day 3	E	x	x	E	G
Day 4	C	D	D	C	X
Day 5	C	x	x	C	C
Day 6	B	B	B	B	X
Day 7	x	D	D	x	X
Day 8	x	H	H	x	X
Day 9	F	F	A	A	G

E could have worked on Day 3 in team T1.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

Eight students - A, B, C, D, E, F, G and H are planning to work on 3 projects - P1, P2, and P3. For these 3 projects they form 3 teams - T1, T2 and T3. Team T1 works on project P1, Team T2 works on project P2 and Team T3 works on project P3. For each project to be successful, the students need to work on certain days, as per the requirement of the projects. The table given below shows information about the different days that the students can work for the projects:

Student	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
A	✓	x	x	x	X	x	x	x	✓
B	X	✓	✓	x	X	✓	x	x	X
C	X	✓	✓	✓	✓	x	✓	x	X
D	X	X	x	✓	✓	x	✓	x	X
E	X	X	✓	X	X	x	✓	x	X
F	✓	✓	x	X	X	x	x	x	✓
G	X	X	✓	X	X	x	x	x	✓
H	X	X	✓	X	X	x	✓	✓	X

Following is the additional information:
(i) The number of students in team T1, T2 and T3 are 5, 5 and 4 respectively. Each of the 8 students is in at least one team and being part of the team, must work for at least one day. No student can work for more than two days for a particular project.
(ii) In all the 3 teams, one student works on Day 1, one student works on Day 2 and one student works on Day 9, such that no two teams have same student who works on Day 9. Also, the student who works on Day 1 for a project does not work on any other day for that particular project.
(iii) F works in team T1 on Day 9 and G is in team T3. C is in two teams. Also, C cannot be in the same team as either H or D.
(iv) If work is done on a project on Day 3, then it will be done on Day 5 also. Similarly, if work is done on a project on Day 8, then it will be done on Day 7 also. If D is in a team, he always works for two days on the project. The same is true for G.
(v) In 2 teams out of 3, work is done on Day 6. Similarly, in 2 teams out of 3, work is done on Day 3 and the same is true for Day 4. While on Day 8, one student of only one team works.
(vi) Project P1 requires the students of Team T1 to work on exactly 7 different days out of the 9 different days listed in the table above. Similarly, project P2 requires students of team T2 to work on exactly 7 different days and project P3 requires students of team T3 to work on exactly 5 different days.

Q.13 [11831809]
Who worked on Day 2 in team T2?

1 ○ C

2 ○ B

3 ○ Either (1) or (2)

4 ◯ F

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Team T3:
From conditions (iii) and (iv), G works on Day 9 as well as on Day 3.
From condition (iv), If work is done on a project on Day 3, then it will be work on Day 5 also. It has to be C as D cannot work on Day 5. Since D cannot work two days in this team
From conditions (i) and (vi), In team T3, there are given to be 4 members and they work on 5 different days with G works on two days. For Day 1 and Day 2, there are two choices for each. If F works on Day 1, then B works on Day 2 and if A works on Day 1, then on Day 2 the work can be done by either F or B.
Step 2:
Team T1 and Team T2:
From condition (ii), one student works on Day 1, one student works on Day 2 and one student works on Day 9 in each team.
From condition (v), in each of the two teams, one student must work on Day 6, one student on Day 4; while in one of the two teams, from condition (iv), one student must work on Day 3 and hence also one student must work on Day 5.
Also, from condition (iv), In the other team, one student must work on Day 8 and hence also one student must work on Day 7.
Step 3:
For teams T1 and T2, either there is case I for both or there is case II for both.
Hence, the final table can be shown as:

	Team T1		Team T2		Team T3
	Case I	Case II	Case I	Case II	
Day 1	A	A	F	F	F/A
Day 2	B/F	B/F	B	B	F/B
Day 3	E	x	x	E	G
Day 4	C	D	D	C	X
Day 5	C	x	x	C	C
Day 6	B	B	B	B	X
Day 7	x	D	D	x	X
Day 8	x	H	H	x	X
Day 9	F	F	A	A	G

B worked on Day 2 in team T2.

Bookmark FeedBack

Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

Eight students - A, B, C, D, E, F, G and H are planning to work on 3 projects - P1, P2, and P3. For these 3 projects they form 3 teams - T1, T2 and T3. Team T1 works on project P1, Team T2 works on project P2 and Team T3 works on project P3. For each project to be successful, the students need to work on certain days, as per the requirement of the projects. The table given below shows information about the different days that the students can work for the projects:

Student	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
A	✓	x	x	x	X	x	x	x	✓
B	X	✓	✓	x	X	✓	x	x	X
C	X	✓	✓	✓	✓	x	✓	x	X
D	X	X	x	✓	✓	x	✓	x	X
E	X	X	✓	X	X	x	✓	x	X
F	✓	✓	x	X	X	x	x	x	✓
G	X	X	✓	X	X	x	x	x	✓
H	X	X	✓	X	X	x	✓	✓	X

Following is the additional information:
(i) The number of students in team T1, T2 and T3 are 5, 5 and 4 respectively. Each of the 8 students is in at least one team and being part of the team, must work for at least one day. No student can work for more than two days for a particular project.
(ii) In all the 3 teams, one student works on Day 1, one student works on Day 2 and one student works on Day 9, such that no two teams have same student who works on Day 9. Also, the student who works on Day 1 for a project does not work on any other day for that particular project.
(iii) F works in team T1 on Day 9 and G is in team T3. C is in two teams. Also, C cannot be in the same team as either H or D.
(iv) If work is done on a project on Day 3, then it will be done on Day 5 also. Similarly, if work is done on a project on Day 8, then it will be done on Day 7 also. If D is in a team, he always works for two days on the project. The same is true for G.
(v) In 2 teams out of 3, work is done on Day 6. Similarly, in 2 teams out of 3, work is done on Day 3 and the same is true for Day 4. While on Day 8, one student of only one team works.
(vi) Project P1 requires the students of Team T1 to work on exactly 7 different days out of the 9 different days listed in the table above. Similarly, project P2 requires students of team T2 to work on exactly 7 different days and project P3 requires students of team T3 to work on exactly 5 different days.

Q.14 [11831809]
Among the following students who could have worked for two days on project P1?

1 ○ F

2 ○ B

3 ○ C

4 ◯ All of these

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Team T3:
From conditions (iii) and (iv), G works on Day 9 as well as on Day 3.
From condition (iv), If work is done on a project on Day 3, then it will be work on Day 5 also. It has to be C as D cannot work on Day 5. Since D cannot work two days in this team
From conditions (i) and (vi), In team T3, there are given to be 4 members and they work on 5 different days with G works on two days. For Day 1 and Day 2, there are two choices for each. If F works on Day 1, then B works on Day 2 and if A works on Day 1, then on Day 2 the work can be done by either F or B.
Step 2:
Team T1 and Team T2:
From condition (ii), one student works on Day 1, one student works on Day 2 and one student works on Day 9 in each team.
From condition (v), in each of the two teams, one student must work on Day 6, one student on Day 4; while in one of the two teams, from condition (iv), one student must work on Day 3 and hence also one student must work on Day 5.
Also, from condition (iv), In the other team, one student must work on Day 8 and hence also one student must work on Day 7.
Step 3:
For teams T1 and T2, either there is case I for both or there is case II for both.
Hence, the final table can be shown as:

	Team T1		Team T2		Team T3
	Case I	Case II	Case I	Case II	
Day 1	A	A	F	F	F/A
Day 2	B/F	B/F	B	B	F/B
Day 3	E	x	x	E	G
Day 4	C	D	D	C	X
Day 5	C	x	x	C	C
Day 6	B	B	B	B	X
Day 7	x	D	D	x	X
Day 8	x	H	H	x	X
Day 9	F	F	A	A	G

All B, C and F could have worked for two days on project P1.

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Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

Eight students - A, B, C, D, E, F, G and H are planning to work on 3 projects - P1, P2, and P3. For these 3 projects they form 3 teams - T1, T2 and T3. Team T1 works on project P1, Team T2 works on project P2 and Team T3 works on project P3. For each project to be successful, the students need to work on certain days, as per the requirement of the projects. The table given below shows information about the different days that the students can work for the projects:

Student	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
A	✓	x	x	x	X	x	x	x	✓
B	X	✓	✓	x	X	✓	x	x	X
C	X	✓	✓	✓	✓	x	✓	x	X
D	X	X	x	✓	✓	x	✓	x	X
E	X	X	✓	X	X	x	✓	x	X
F	✓	✓	x	X	X	x	x	x	✓
G	X	X	✓	X	X	x	x	x	✓
H	X	X	✓	X	X	x	✓	✓	X

Following is the additional information:
(i) The number of students in team T1, T2 and T3 are 5, 5 and 4 respectively. Each of the 8 students is in at least one team and being part of the team, must work for at least one day. No student can work for more than two days for a particular project.
(ii) In all the 3 teams, one student works on Day 1, one student works on Day 2 and one student works on Day 9, such that no two teams have same student who works on Day 9. Also, the student who works on Day 1 for a project does not work on any other day for that particular project.
(iii) F works in team T1 on Day 9 and G is in team T3. C is in two teams. Also, C cannot be in the same team as either H or D.
(iv) If work is done on a project on Day 3, then it will be done on Day 5 also. Similarly, if work is done on a project on Day 8, then it will be done on Day 7 also. If D is in a team, he always works for two days on the project. The same is true for G.
(v) In 2 teams out of 3, work is done on Day 6. Similarly, in 2 teams out of 3, work is done on Day 3 and the same is true for Day 4. While on Day 8, one student of only one team works.
(vi) Project P1 requires the students of Team T1 to work on exactly 7 different days out of the 9 different days listed in the table above. Similarly, project P2 requires students of team T2 to work on exactly 7 different days and project P3 requires students of team T3 to work on exactly 5 different days.

Q.15 [11831809]
Which of the following students cannot work on project P2?

1 ○ A

2 ○ D

3 ○ G

4 ◯ H

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Team T3:
From conditions (iii) and (iv), G works on Day 9 as well as on Day 3.
From condition (iv), If work is done on a project on Day 3, then it will be work on Day 5 also. It has to be C as D cannot work on Day 5. Since D cannot work two days in this team
From conditions (i) and (vi), In team T3, there are given to be 4 members and they work on 5 different days with G works on two days. For Day 1 and Day 2, there are two choices for each. If F works on Day 1, then B works on Day 2 and if A works on Day 1, then on Day 2 the work can be done by either F or B.
Step 2:
Team T1 and Team T2:
From condition (ii), one student works on Day 1, one student works on Day 2 and one student works on Day 9 in each team.
From condition (v), in each of the two teams, one student must work on Day 6, one student on Day 4; while in one of the two teams, from condition (iv), one student must work on Day 3 and hence also one student must work on Day 5.
Also, from condition (iv), In the other team, one student must work on Day 8 and hence also one student must work on Day 7.
Step 3:
For teams T1 and T2, either there is case I for both or there is case II for both.
Hence, the final table can be shown as:

	Team T1		Team T2		Team T3
	Case I	Case II	Case I	Case II	
Day 1	A	A	F	F	F/A
Day 2	B/F	B/F	B	B	F/B
Day 3	E	x	x	E	G
Day 4	C	D	D	C	X
Day 5	C	x	x	C	C
Day 6	B	B	B	B	X
Day 7	x	D	D	x	X
Day 8	x	H	H	x	X
Day 9	F	F	A	A	G

G cannot work on project P2.

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Directions for questions 11 to 16: Answer the questions on the basis of the information given below.

Eight students - A, B, C, D, E, F, G and H are planning to work on 3 projects - P1, P2, and P3. For these 3 projects they form 3 teams - T1, T2 and T3. Team T1 works on project P1, Team T2 works on project P2 and Team T3 works on project P3. For each project to be successful, the students need to work on certain days, as per the requirement of the projects. The table given below shows information about the different days that the students can work for the projects:

Student	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
A	✓	x	x	x	X	x	x	x	✓
B	X	✓	✓	x	X	✓	x	x	X
C	X	✓	✓	✓	✓	x	✓	x	X
D	X	X	x	✓	✓	x	✓	x	X
E	X	X	✓	X	X	x	✓	x	X
F	✓	✓	x	X	X	x	x	x	✓
G	X	X	✓	X	X	x	x	x	✓
H	X	X	✓	X	X	x	✓	✓	X

Following is the additional information:
(i) The number of students in team T1, T2 and T3 are 5, 5 and 4 respectively. Each of the 8 students is in at least one team and being part of the team, must work for at least one day. No student can work for more than two days for a particular project.
(ii) In all the 3 teams, one student works on Day 1, one student works on Day 2 and one student works on Day 9, such that no two teams have same student who works on Day 9. Also, the student who works on Day 1 for a project does not work on any other day for that particular project.
(iii) F works in team T1 on Day 9 and G is in team T3. C is in two teams. Also, C cannot be in the same team as either H or D.
(iv) If work is done on a project on Day 3, then it will be done on Day 5 also. Similarly, if work is done on a project on Day 8, then it will be done on Day 7 also. If D is in a team, he always works for two days on the project. The same is true for G.
(v) In 2 teams out of 3, work is done on Day 6. Similarly, in 2 teams out of 3, work is done on Day 3 and the same is true for Day 4. While on Day 8, one student of only one team works.
(vi) Project P1 requires the students of Team T1 to work on exactly 7 different days out of the 9 different days listed in the table above. Similarly, project P2 requires students of team T2 to work on exactly 7 different days and project P3 requires students of team T3 to work on exactly 5 different days.

Q.16 [11831809]
In which of the following days maximum number of students work?

1 ○ Day 2

2 ○ Day 3

3 ○ Day 4

4 ◯ Day 9

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Team T3:
From conditions (iii) and (iv), G works on Day 9 as well as on Day 3.
From condition (iv), If work is done on a project on Day 3, then it will be work on Day 5 also. It has to be C as D cannot work on Day 5. Since D cannot work two days in this team
From conditions (i) and (vi), In team T3, there are given to be 4 members and they work on 5 different days with G works on two days. For Day 1 and Day 2, there are two choices for each. If F works on Day 1, then B works on Day 2 and if A works on Day 1, then on Day 2 the work can be done by either F or B.
Step 2:
Team T1 and Team T2:
From condition (ii), one student works on Day 1, one student works on Day 2 and one student works on Day 9 in each team.
From condition (v), in each of the two teams, one student must work on Day 6, one student on Day 4; while in one of the two teams, from condition (iv), one student must work on Day 3 and hence also one student must work on Day 5.
Also, from condition (iv), In the other team, one student must work on Day 8 and hence also one student must work on Day 7.
Step 3:
For teams T1 and T2, either there is case I for both or there is case II for both.
Hence, the final table can be shown as:

	Team T1		Team T2		Team T3
	Case I	Case II	Case I	Case II	
Day 1	A	A	F	F	F/A
Day 2	B/F	B/F	B	B	F/B
Day 3	E	x	x	E	G
Day 4	C	D	D	C	X
Day 5	C	x	x	C	C
Day 6	B	B	B	B	X
Day 7	x	D	D	x	X
Day 8	x	H	H	x	X
Day 9	F	F	A	A	G

On Day 9 maximum 3 students A, F and G work.

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Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

In the 5 × 5 grid given below, 25 consecutive integers starting from x are to be arranged such that each integer, except x and (x + 1), can be written as the sum of its some two neighbours. Some of the integers are already filled as disguised. Integers in the grid are neighbours if their cells touch along a side or at a vertex. Each integer is used exactly once. For example, (x + 8) and $((x + 4)^2 - 1)$ are neighbours of $(x + 4)^2$. Columns 1st, 2nd, …, 5th are from left to right and rows 1st, 2nd, …, 5th are from top to bottom.

			$(x + 4)^2 - 5$	$(x + 4)^2 - 4$
	(x + 5)	(x + 4)	(x + 3)	
$(x + 4)^2 - 2$	(x + 6)	x	(x + 2)	n
	(x + 8)	(x + 7)	(x + 1)	
$(x + 4)^2$	$(x + 4)^2 - 1$			$(x + 4)^2 - 3$

Q.17 [11831809]

How many different value(s) of x is/are possible?

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:

The possible values of x such that x and $(x + 4)^2$ both fall within 25 consecutive integers are from –8 to 1.

a	b	c	20	21
d	6	5	4	e
23	7	1	3	n
f	9	8	2	g
25	24	h	i	22

x is the smallest integer. (x + 1) is the next integer and (x + 2) will be the next integer.
So, (x + 2) can only be written as sum of x and (x + 1).
Every other integer apart from these three will never add up to give (x + 2).
So, x + (x + 1) = (x + 2)
$\Rightarrow x = 1$.

Step 2:

The grid will look like the following.
Let's fill the cells with missing integers in the grid now with letters a through i.
Let's start from top right corner (21).
Since it should be sum of any two from (20, 4 and e), 20 + e will give e = 1 which is already used.
And 20 + 4 is 24 ≠ 21. So, 4 + e = 21 gives e = 17.
Let's now move to bottom left corner (25).
Since it should be sum of any two from (f, 9 and 24), f + 24 will give f = 1 which is already used.
And 9 + 24 = 33 ≠ 25. So, f + 9 = 25 gives f = 16.
Let's now move to bottom right corner (22).
Since it should be sum of any two from (2, g and i), 2 + g or l = 22 gives g or l = 20 which is already used (top row, second from right). So, g + i = 22.
The only combination which makes with unused integers is 10 + 12.
Either g = 10 and i = 12 or g = 12 and i = 10.
If g = 10, it cannot be written as sum of any two of its neighbours (n ≠ 7, 8 or 10; 3; 2; 12 and 22).
So, the only possibility is i = 10 and g = 12. The grid so far looks like this.

a	b	c	20	21
d	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 3:

Unused integers so far are 11, 13, 14, 15, 18, 19 to be filled at a, b, c, d, h and n.
Now, n can be either 14, 15 or 19 only. Also, h can be 11, 18 or 19. So, let's check a, b, c and d. The top left corner (a) cannot be sum of b and d as it will make a more than 20 irrespective of whichever values we put at b and d (since each is in 10s). So, a should be sum of 6 and b or d. Which means a should 6 more than b or d. That is only possible if a is 19 and b or d is 13 because no other unused integers gave a difference of 6. So, a is definitely 19. If b = 13, d has to be sum of any two out of a (=19), b (=13), 6, 7 and 23. No such sum will be formed which is either left or have not already formed. So, b ≠ 13. Definitely d = 13 (sum of 6 and 7). The grid so far looks like this.

19	b	c	20	21
13	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 4:

Unused integers so far are 11, 14, 15 and 18 to be filled at b, c, h and n.
The only possibility is b is 11 (= 5 + 6), c is 15 (= 11 + 4), h is 18 (= 10 + 8) and n is 14 (= 2 + 12). No other combination will fulfil all the conditions.
The final grid is as follows:

19	11	15	20	21
13	6	5	4	17
23	7	1	3	14

16	9	8	2	12
25	24	18	10	22

The value of x is only 1.

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Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

In the 5 × 5 grid given below, 25 consecutive integers starting from x are to be arranged such that each integer, except x and (x + 1), can be written as the sum of its some two neighbours. Some of the integers are already filled as disguised. Integers in the grid are neighbours if their cells touch along a side or at a vertex. Each integer is used exactly once. For example, (x + 8) and $((x + 4)^2 - 1)$ are neighbours of $(x + 4)^2$. Columns 1st, 2nd, …, 5th are from left to right and rows 1st, 2nd, …, 5th are from top to bottom.

			$(x + 4)^2 - 5$	$(x + 4)^2 - 4$
	(x + 5)	(x + 4)	(x + 3)	
$(x + 4)^2 - 2$	(x + 6)	x	(x + 2)	n
	(x + 8)	(x + 7)	(x + 1)	
$(x + 4)^2$	$(x + 4)^2 - 1$			$(x + 4)^2 - 3$

Q.18 [11831809]

What is the average of all the integers in the 1st column cells?

1 ◯ 18.2

2 ◯ 19

3 ◯ 19.2

4 ◯ Either (1) or (3)

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:

The possible values of x such that x and $(x + 4)^2$ both fall within 25 consecutive integers are from –8 to 1.

a	b	c	20	21
d	6	5	4	e
23	7	1	3	n
f	9	8	2	g
25	24	h	i	22

x is the smallest integer. (x + 1) is the next integer and (x + 2) will be the next integer.
So, (x + 2) can only be written as sum of x and (x + 1).
Every other integer apart from these three will never add up to give (x + 2).
So, x + (x + 1) = (x + 2)
$\Rightarrow x = 1$.

Step 2:

The grid will look like the following.
Let's fill the cells with missing integers in the grid now with letters a through i.
Let's start from top right corner (21).
Since it should be sum of any two from (20, 4 and e), 20 + e will give e = 1 which is already used.
And 20 + 4 is 24 ≠ 21. So, 4 + e = 21 gives e = 17.
Let's now move to bottom left corner (25).
Since it should be sum of any two from (f, 9 and 24), f + 24 will give f = 1 which is already used.
And 9 + 24 = 33 ≠ 25. So, f + 9 = 25 gives f = 16.
Let's now move to bottom right corner (22).
Since it should be sum of any two from (2, g and i), 2 + g or l = 22 gives g or l = 20 which is already used (top row, second from right). So, g + i = 22.
The only combination which makes with unused integers is 10 + 12.
Either g = 10 and i = 12 or g = 12 and i = 10.
If g = 10, it cannot be written as sum of any two of its neighbours (n ≠ 7, 8 or 10; 3; 2; 12 and 22).
So, the only possibility is i = 10 and g = 12. The grid so far looks like this.

a	b	c	20	21
d	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 3:

Unused integers so far are 11, 13, 14, 15, 18, 19 to be filled at a, b, c, d, h and n.
Now, n can be either 14, 15 or 19 only. Also, h can be 11, 18 or 19. So, let's check a, b, c and d. The top left corner (a) cannot be sum of b and d as it will make a more than 20 irrespective of whichever values we put at b and d (since each is in 10s). So, a should be sum of 6 and b or d. Which means a should 6 more than b or d. That is only possible if a is 19 and b or d is 13 because no other unused integers gave a difference of 6. So, a is definitely 19. If b = 13, d has to be sum of any two out of a (=19), b (=13), 6, 7 and 23. No such sum will be formed which is either left or have not already formed. So, b ≠ 13. Definitely d = 13 (sum of 6 and 7). The grid so far looks like this.

19	b	c	20	21
13	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 4:

Unused integers so far are 11, 14, 15 and 18 to be filled at b, c, h and n.
The only possibility is b is 11 (= 5 + 6), c is 15 (= 11 + 4), h is 18 (= 10 + 8) and n is 14 (= 2 + 12). No other combination will fulfil all the conditions.
The final grid is as follows:

19	11	15	20	21
13	6	5	4	17
23	7	1	3	14

16	9	8	2	12
25	24	18	10	22

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Required average = (19 + 13 + 23 + 16 + 25)/5 = 96/5 = 19.2.

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

In the 5 × 5 grid given below, 25 consecutive integers starting from x are to be arranged such that each integer, except x and (x + 1), can be written as the sum of its some two neighbours. Some of the integers are already filled as disguised. Integers in the grid are neighbours if their cells touch along a side or at a vertex. Each integer is used exactly once. For example, (x + 8) and $((x + 4)^2 - 1)$ are neighbours of $(x + 4)^2$. Columns 1st, 2nd, …, 5th are from left to right and rows 1st, 2nd, …, 5th are from top to bottom.

			$(x + 4)^2 - 5$	$(x + 4)^2 - 4$
	(x + 5)	(x + 4)	(x + 3)	
$(x + 4)^2 - 2$	(x + 6)	x	(x + 2)	n
	(x + 8)	(x + 7)	(x + 1)	
$(x + 4)^2$	$(x + 4)^2 - 1$			$(x + 4)^2 - 3$

Q.19 [11831809]

What is the sum of all the integers in the primary diagonal cells (cells from top left to bottom right)?

1 ◯ 45

2 ◯ 50

3 ◯ 55

4 ◯ 60

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:

The possible values of x such that x and $(x + 4)^2$ both fall within 25 consecutive integers are from –8 to 1.

a	b	c	20	21
d	6	5	4	e
23	7	1	3	n
f	9	8	2	g
25	24	h	i	22

x is the smallest integer. (x + 1) is the next integer and (x + 2) will be the next integer.
So, (x + 2) can only be written as sum of x and (x + 1).
Every other integer apart from these three will never add up to give (x + 2).
So, x + (x + 1) = (x + 2)
$\Rightarrow x = 1$.

Step 2:

The grid will look like the following.
Let's fill the cells with missing integers in the grid now with letters a through i.
Let's start from top right corner (21).
Since it should be sum of any two from (20, 4 and e), 20 + e will give e = 1 which is already used.
And 20 + 4 is 24 ≠ 21. So, 4 + e = 21 gives e = 17.
Let's now move to bottom left corner (25).
Since it should be sum of any two from (f, 9 and 24), f + 24 will give f = 1 which is already used.
And 9 + 24 = 33 ≠ 25. So, f + 9 = 25 gives f = 16.
Let's now move to bottom right corner (22).
Since it should be sum of any two from (2, g and i), 2 + g or l = 22 gives g or l = 20 which is already used (top row, second from right). So, g + i = 22.
The only combination which makes with unused integers is 10 + 12.
Either g = 10 and i = 12 or g = 12 and i = 10.
If g = 10, it cannot be written as sum of any two of its neighbours (n ≠ 7, 8 or 10; 3; 2; 12 and 22).
So, the only possibility is i = 10 and g = 12. The grid so far looks like this.

a	b	c	20	21
d	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 3:

Unused integers so far are 11, 13, 14, 15, 18, 19 to be filled at a, b, c, d, h and n.
Now, n can be either 14, 15 or 19 only. Also, h can be 11, 18 or 19. So, let's check a, b, c and d. The top left corner (a) cannot be sum of b and d as it will make a more than 20 irrespective of whichever values we put at b and d (since each is in 10s). So, a should be sum of 6 and b or d. Which means a should 6 more than b or d. That is only possible if a is 19 and b or d is 13 because no other unused integers gave a difference of 6. So, a is definitely 19. If b = 13, d has to be sum of any two out of a (=19), b (=13), 6, 7 and 23. No such sum will be formed which is either left or have not already formed. So, b ≠ 13. Definitely d = 13 (sum of 6 and 7). The grid so far looks like this.

19	b	c	20	21
13	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 4:

Unused integers so far are 11, 14, 15 and 18 to be filled at b, c, h and n.
The only possibility is b is 11 (= 5 + 6), c is 15 (= 11 + 4), h is 18 (= 10 + 8) and n is 14 (= 2 + 12). No other combination will fulfil all the conditions.
The final grid is as follows:

19	11	15	20	21
13	6	5	4	17
23	7	1	3	14

16	9	8	2	12
25	24	18	10	22

Required sum = 19 + 6 + 1 + 2 + 22 = 50.

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Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

In the 5 × 5 grid given below, 25 consecutive integers starting from x are to be arranged such that each integer, except x and (x + 1), can be written as the sum of its some two neighbours. Some of the integers are already filled as disguised. Integers in the grid are neighbours if their cells touch along a side or at a vertex. Each integer is used exactly once. For example, (x + 8) and $((x + 4)^2 - 1)$ are neighbours of $(x + 4)^2$. Columns 1st, 2nd, …, 5th are from left to right and rows 1st, 2nd, …, 5th are from top to bottom.

			$(x + 4)^2 - 5$	$(x + 4)^2 - 4$
	(x + 5)	(x + 4)	(x + 3)	
$(x + 4)^2 - 2$	(x + 6)	x	(x + 2)	n
	(x + 8)	(x + 7)	(x + 1)	
$(x + 4)^2$	$(x + 4)^2 - 1$			$(x + 4)^2 - 3$

Q.20 [11831809]

Which number is “n” in the grid?

1 ◯ 11

2 ◯ 14

3 ◯ 19

4 ◯ Either (2) or (3)

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:

The possible values of x such that x and $(x + 4)^2$ both fall within 25 consecutive integers are from –8 to 1.

a	b	c	20	21
d	6	5	4	e
23	7	1	3	n
f	9	8	2	g
25	24	h	i	22

x is the smallest integer. (x + 1) is the next integer and (x + 2) will be the next integer.
So, (x + 2) can only be written as sum of x and (x + 1).
Every other integer apart from these three will never add up to give (x + 2).
So, x + (x + 1) = (x + 2)
$\Rightarrow x = 1$.

Step 2:

The grid will look like the following.
Let's fill the cells with missing integers in the grid now with letters a through i.
Let's start from top right corner (21).
Since it should be sum of any two from (20, 4 and e), 20 + e will give e = 1 which is already used.
And 20 + 4 is 24 ≠ 21. So, 4 + e = 21 gives e = 17.
Let's now move to bottom left corner (25).
Since it should be sum of any two from (f, 9 and 24), f + 24 will give f = 1 which is already used.
And 9 + 24 = 33 ≠ 25. So, f + 9 = 25 gives f = 16.
Let's now move to bottom right corner (22).
Since it should be sum of any two from (2, g and i), 2 + g or l = 22 gives g or l = 20 which is already used (top row, second from right). So, g + i = 22.
The only combination which makes with unused integers is 10 + 12.
Either g = 10 and i = 12 or g = 12 and i = 10.
If g = 10, it cannot be written as sum of any two of its neighbours (n ≠ 7, 8 or 10; 3; 2; 12 and 22).
So, the only possibility is i = 10 and g = 12. The grid so far looks like this.

a	b	c	20	21
d	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 3:

Unused integers so far are 11, 13, 14, 15, 18, 19 to be filled at a, b, c, d, h and n.
Now, n can be either 14, 15 or 19 only. Also, h can be 11, 18 or 19. So, let's check a, b, c and d. The top left corner (a) cannot be sum of b and d as it will make a more than 20 irrespective of whichever values we put at b and d (since each is in 10s). So, a should be sum of 6 and b or d. Which means a should 6 more than b or d. That is only possible if a is 19 and b or d is 13 because no other unused integers gave a difference of 6. So, a is definitely 19. If b = 13, d has to be sum of any two out of a (=19), b (=13), 6, 7 and 23. No such sum will be formed which is either left or have not already formed. So, b ≠ 13. Definitely d = 13 (sum of 6 and 7). The grid so far looks like this.

19	b	c	20	21
13	6	5	4	17
23	7	1	3	n
16	9	8	2	12
25	24	h	10	22

Step 4:

Unused integers so far are 11, 14, 15 and 18 to be filled at b, c, h and n.
The only possibility is b is 11 (= 5 + 6), c is 15 (= 11 + 4), h is 18 (= 10 + 8) and n is 14 (= 2 + 12). No other combination will fulfil all the conditions.
The final grid is as follows:

19	11	15	20	21
13	6	5	4	17
23	7	1	3	14

16	9	8	2	12
25	24	18	10	22

The number n is 14.

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